

ITF23019: Parallel and Distributed Programming

Report for Lab 12: Practice with Cloud Computing

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Exercise 1:

Prerequisites: Before I created the files and followed the steps as described, I made a folder-structure in docker “root/ITI42929/new_folder” as shown in guide.

Part 1: Windows to Docker

Subtask 1: Create *windows_file.txt* in Windows

- I created this file on Windows desktop in Notepad.

Subtask 2: Copy file from Windows to Docker

- I did this by using the provided command, and copied the file from Windows desktop to the folder in Docker with path root/ITI42920/new_folder

```
PS C:\Users\ITF23019\Desktop> docker cp C:\Users\ITF23019\Desktop\windows_file.txt sandbox-hdp:/root/ITI42920/new_folder/windows_file.txt
Successfully copied 2.05kB to sandbox-hdp:/root/ITI42920/new_folder/windows_file.txt
PS C:\Users\ITF23019\Desktop> _
```

Subtask 3: Confirm that the file was copied over

```
[root@sandbox-hdp new_folder]# pwd
/root/ITI42920/new_folder
[root@sandbox-hdp new_folder]# ls
windows_file.txt
[root@sandbox-hdp new_folder]# █
```

Part 2: Docker to Windows

Subtask 1: Create *linux_file.txt* in terminal

- I created the file with the provided cat command

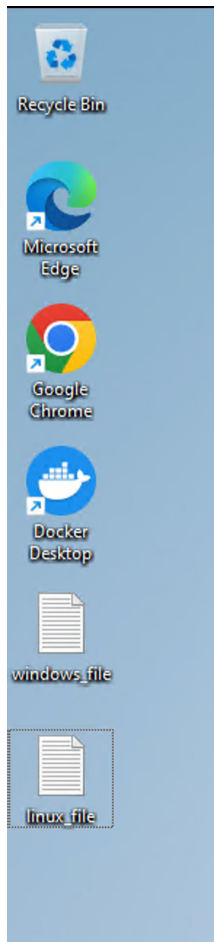
```
[root@sandbox-hdp new_folder]# cat >linux_file.txt
This is a .txt file created in Linux.
^C
[root@sandbox-hdp new_folder]# cat linux_file.txt
This is a .txt file created in Linux.
[root@sandbox-hdp new_folder]# █
```

Subtask 2: Copy file from Docker to Windows

```
PS C:\Users\ITF23019> docker cp sandbox-hdp:/root/ITI42920/new_folder/linux_file.txt C:\Users\ITF23019\Desktop\linux_file.txt
Successfully copied 2.05kB to C:\Users\ITF23019\Desktop\linux_file.txt
PS C:\Users\ITF23019> _
```

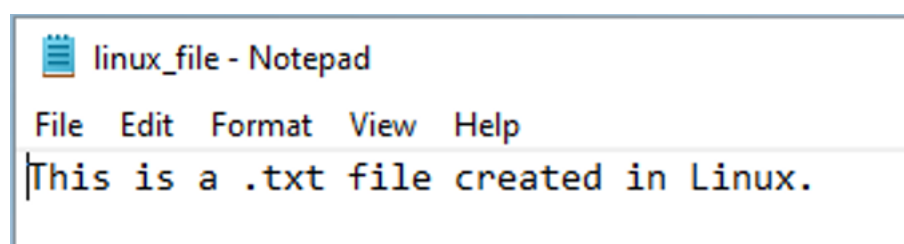
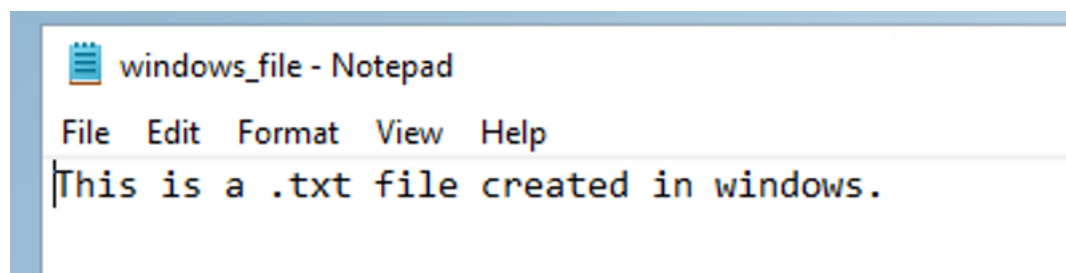
Subtask 3: Confirm that the file was copied over

- Under are two pictures showing that both files are in fact on the Windows Desktop and in Docker. This means that the copying went as expected, both ways.



```
[root@sandbox-hdp new_folder]# pwd
/root/ITI42920/new_folder
[root@sandbox-hdp new_folder]# ls -l
total 8
-rw-r--r-- 1 root root 38 Apr 13 21:09 linux_file.txt
-rwxr-xr-x 1 root root 39 Apr 13 20:46 windows_file.txt
[root@sandbox-hdp new_folder]#
```

Content of both files:



Exercise 2:

Subtask 1: What virtual resource do you have in this lab?

- In this lab we have an Azure Labs Virtual Machine. This is a cloud-hosted Virtual Machine (VM), which has the following specs:
 - o 4 CPUs
 - o 16 GB of RAM
 - o 128 GB Premium SSD

Subtask 2: What type of service are you using in this lab?

- In this lab we are using IaaS – Infrastructure as a Service. We are provided with the “raw infrastructure resources” such as a virtual machine. We have direct control over the OS and can manually change the software as we like, such as Docker, networking and file operations. The university is also billed on a “pay-as-you-go” pricing model, which is typical for an IaaS cloud service.

Subtask 3: Where is the user interface that you used to access cloud services?

- The user interface used to access the cloud service is the Azure Labs web portal. This is accessed through a web browser using the provided link sent to our school email. Here we can start a VM and connect to it.

Subtask 4: How responsive is the Azure Labs service to your operations?

- The Azure Labs service was very responsive and smooth to work with throughout the lab. Operations such as starting the VM, connecting via RDP, and running commands in PowerShell and the Shell Web Client all completed quickly and without any major issues.